DATA SCIENCE & ANALYTICS · SESSION 04

Exploratory Data Analysis

Letting the data show you its shape before you commit to any conclusion

Asst.Prof.Dr. Chaiwat N. · 2026

2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

Session Agenda

01 **What EDA Is — and Why**
Exploration before confirmation; the question-first mindset

02 **Distributions**
Histograms, boxplots, skewness, and kurtosis

03 **Correlations**
.corr(), heatmaps, and scatter matrices

04 **Outliers**
The IQR method, z-scores, and visual spotting

05 **From Patterns to Hypotheses**
Turning what you see into questions worth testing

06 **Workshop Block**
Run a structured EDA on a dataset — graded

01

PART 01

What EDA Is — and Why

Exploratory data analysis is the open-minded first look: you let the data reveal its distributions, relationships, and surprises before you test anything.

EDA is detective work: look first, theorize second, confirm last

The mindset

Explore vs confirm

Confirmatory analysis tests a fixed hypothesis. EDA comes first — it generates the hypotheses worth testing at all.

Question-first

Start with a question ('which students struggle?'), then let plots suggest where the answer hides. Don't plot aimlessly.

Why it matters

Skipping EDA means modeling data you don't understand — outliers, skew, and broken columns sink the model silently.

The ritual

A repeatable order: distributions, then correlations, then outliers, then hypotheses. The same four moves every time.

Source: DS Course, Session 4, 2026

02

PART 02

Distributions

Before comparing or modeling anything, understand the shape of each variable on its own — its center, spread, skew, and tails.

Histograms & boxplots: the shape of a single variable at a glance

See one column's shape

Histogram

Bins values into bars — shows where data clusters, whether it's one peak or several, and how wide the spread is.

Boxplot

Shows median, quartiles, and outliers in one compact picture. Ideal for comparing a number across categories.

Read it

Is it symmetric or lopsided? One hump or two? A long tail? The shape decides which statistics are even meaningful.

PYTHON · DISTRIBUTIONS

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.histplot(df['gpa'], bins=20, kde=True)
plt.show()

# compare a number across groups
sns.boxplot(data=df, x='faculty', y='gpa')
plt.show()
```

Always look at the distribution before the mean.

Skewness and kurtosis: putting a number on the shape

Quantify the tails

Skewness	Measures lopsidedness. Positive = a long right tail (a few big values); zero = symmetric. Income and prices skew right.
Kurtosis	Measures tail-heaviness. High kurtosis = more extreme outliers than a normal curve — fat tails that risk models ignore.
Why the mean lies	On skewed data the mean is dragged toward the tail. Report the median too, or you'll misstate the 'typical' value.
So what	Skew and fat tails decide whether you transform a variable, pick a robust statistic, or treat outliers specially.

Source: DS Course, Session 4, 2026

03

PART 03

Correlations

Once you know each variable alone, look at how they move together — the relationships that hint at drivers, redundancy, and stories.

Correlation: which variables move together, and how strongly

Relationships at a glance

.corr()

Returns a matrix of pairwise correlations, -1 to +1. Near +1 move together, near -1 move oppositely, near 0 unrelated.

Heatmap

Colour the matrix so strong relationships pop out instantly. The fastest way to scan dozens of variables at once.

Correlation != cause

Two things moving together may share a hidden third cause. Correlation flags a lead to investigate, not a conclusion.

PYTHON · CORRELATION

```
corr = df.corr(numeric_only=True)

import seaborn as sns
sns.heatmap(corr, annot=True,
            cmap='coolwarm', center=0)
plt.show()

# pairwise scatter for a quick scan
sns.pairplot(df[['gpa','study_hours','age']])
```

A heatmap turns a number grid into an instant read.

04

PART 04

Outliers

Extreme values can be the most important signal or a data error — either way you must find them deliberately, not let them quietly distort everything.

Two ways to flag outliers: the IQR fence and the z-score

Find the extremes

IQR method

Anything below Q1 − 1.5·IQR or above Q3 + 1.5·IQR is flagged. Robust and distribution-free — the boxplot's logic in code.

Z-score

How many standard deviations from the mean. |z| > 3 is a common cutoff — but it assumes a roughly normal column.

Then decide

Flagging isn't deleting. Investigate: a data error gets fixed, a real extreme gets kept and understood.

PYTHON · OUTLIERS

```
q1, q3 = df['gpa'].quantile([.25, .75])
iqr = q3 - q1
lo, hi = q1 - 1.5*iqr, q3 + 1.5*iqr
outliers = df[(df['gpa'] < lo) |
              (df['gpa'] > hi)]

# z-score alternative
z = (df['gpa'] - df['gpa'].mean()) / df['gpa'].std()
df[z.abs() > 3]
```

Flag, investigate, then decide — never delete on sight.

05

PART 05

From Patterns to Hypotheses

EDA's real output isn't charts — it's a short list of sharp, testable hypotheses that focus everything you do next.

The point of EDA is the questions it produces, not the plots it makes

Turn observations into tests

Observe

'Engineering GPAs cluster lower and study hours are higher.' A concrete pattern you can see in a chart.

Hypothesize

Phrase it as something testable: 'higher study hours don't lift GPA in Engineering.' Now it has a yes/no answer.

Prioritize

Rank hypotheses by impact and how easily the data can answer them. Chase the few that matter, not every curiosity.

Hand off

These hypotheses become your modeling targets and your story's spine. EDA feeds Sessions 5 through 12.

Source: DS Course, Session 4, 2026

06

PART 06

Workshop Block

Run the full EDA ritual on a dataset — distributions, correlations, outliers — and write three hypotheses the data suggests. Graded.

By the end of this block, every student has run a structured EDA and stated three hypotheses

4 Steps · 35 Minutes

Distributions	Plot a histogram and boxplot for each key numeric column. Note skew and anything bimodal or surprising.
Correlations	Build the correlation heatmap. Mark the two strongest relationships and one that surprises you.
Outliers	Apply the IQR fence to one column. Decide, with a reason, whether each flagged point stays or goes.
Three hypotheses	Write three testable statements the data suggests. This list is your graded deliverable.

Source: Workshop, Session 4, 2026

KEY TAKEAWAY

Look before you model. The data will tell you what to ask.